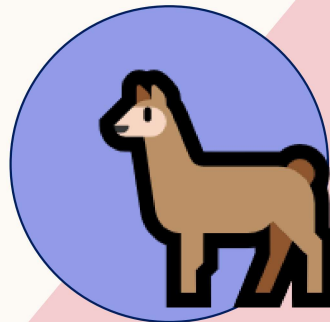


Introduction to Building LLMs

CS229: Machine Learning

Yann Dubois | Aug. 13th 2024

Slides partially based on CS336, CS224N, CS324



LLMs

- LLMs & chatbots took over the world



- How do they work?

ChatGPT



What are you?

I'm a large language model trained by OpenAI. I'm a form of artificial intelligence that has been designed to process and generate human-like language.



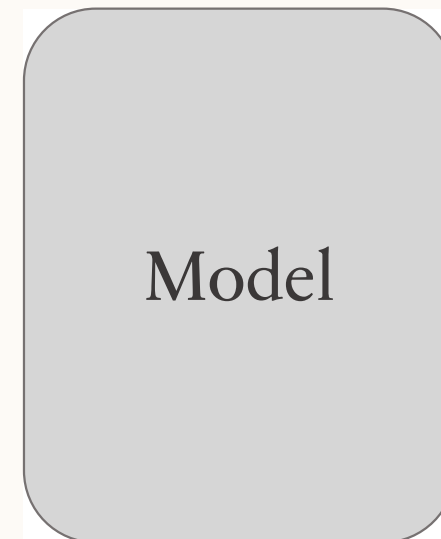
Are you human?

I'm not a human and I don't have the ability to think or feel in the same way that a person does.



What matters when training LLMs

- Architecture
 - Training algorithm/loss
- } Most of academia
- Data
 - Evaluation
 - Systems
- } What matters in practice

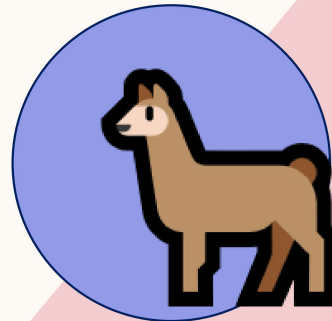


Overview

Pretraining -> GPT3

- Task & loss

Post-training -> ChatGPT



Language Modeling

- LM: probability distribution over sequences of tokens/words $p(x_1, \dots, x_L)$

$$P(\text{the, mouse, ate, the, cheese}) = 0.02$$

$$P(\text{the, the, mouse, ate, cheese}) = 0.0001 \quad \text{Syntactic knowledge}$$

$$P(\text{the, cheese, ate, the, mouse}) = 0.001 \quad \text{Semantic knowledge}$$

- LMs are generative models: $x_{1:L} \sim p(x_1, \dots, x_L)$

- Autoregressive (AR) language models:**

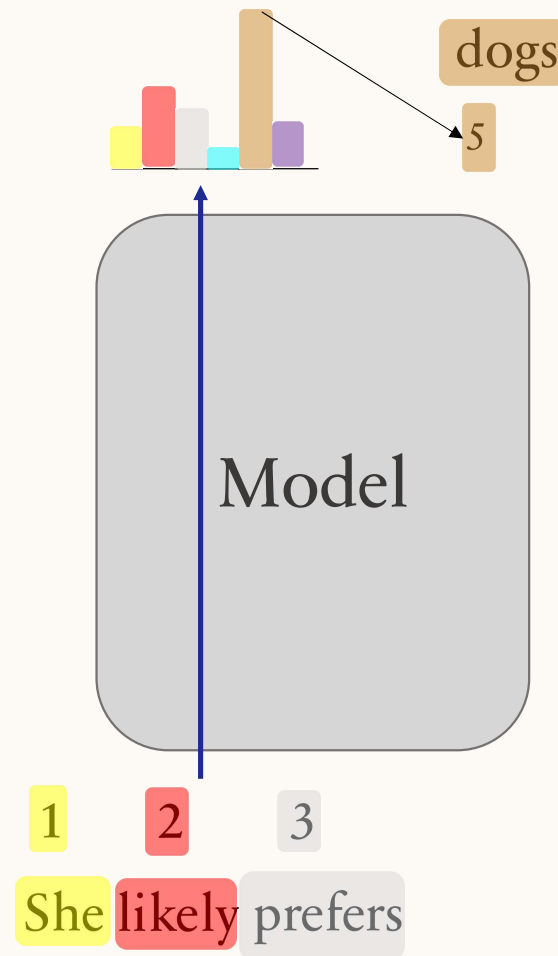
$$p(x_1, \dots, x_L) = p(x_1)p(x_2|x_1)p(x_3|x_2, x_1)\dots = \prod_i p(x_i|x_{1:i-1})$$

No approx: chain rule of probability

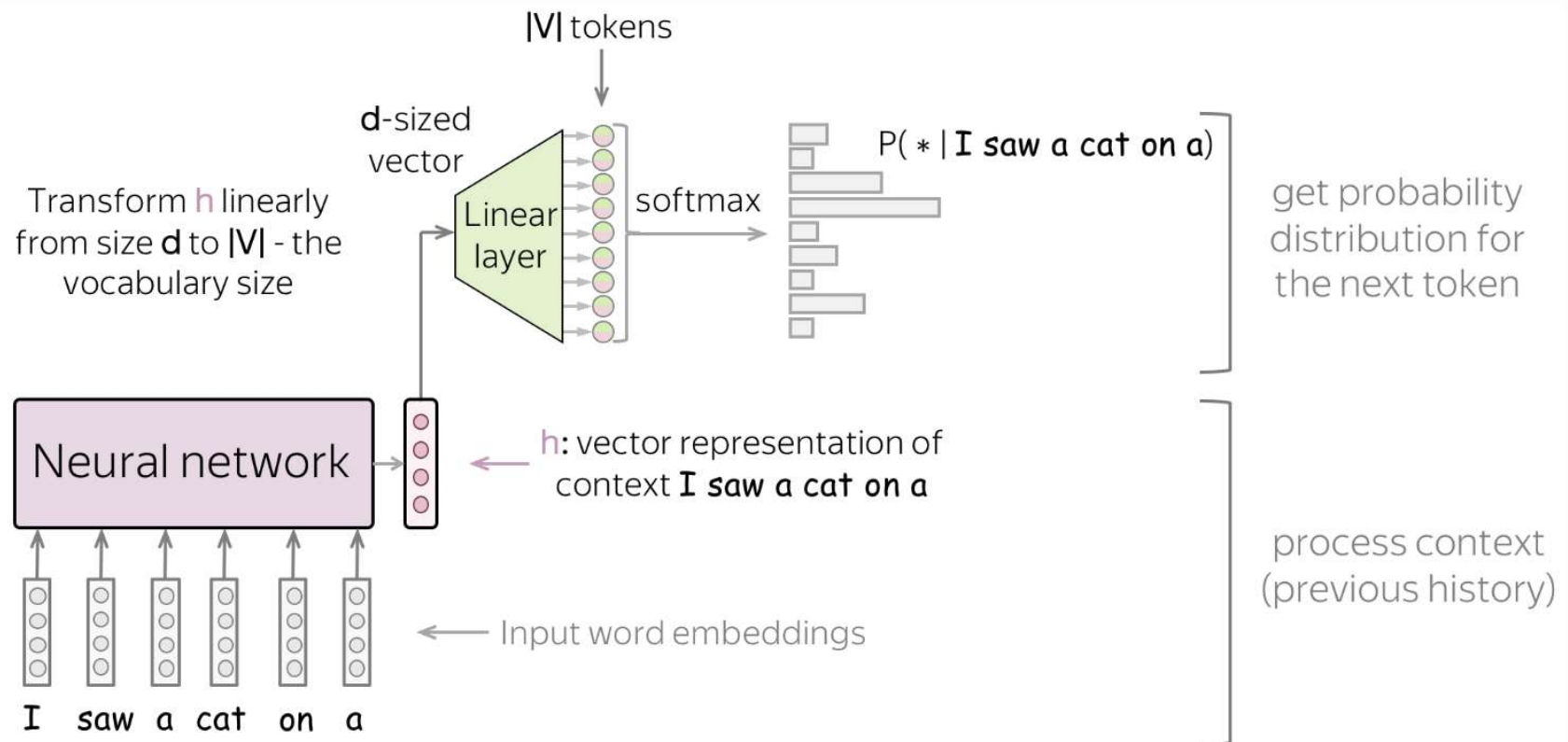
=> You only need a model that can predict the next token given past context!

AR Language Models

- Task: predict the next word
 - Steps:
 1. tokenize
 2. forward
 3. predict probability of next token
 4. sample
 5. detokenize
- } Inference only

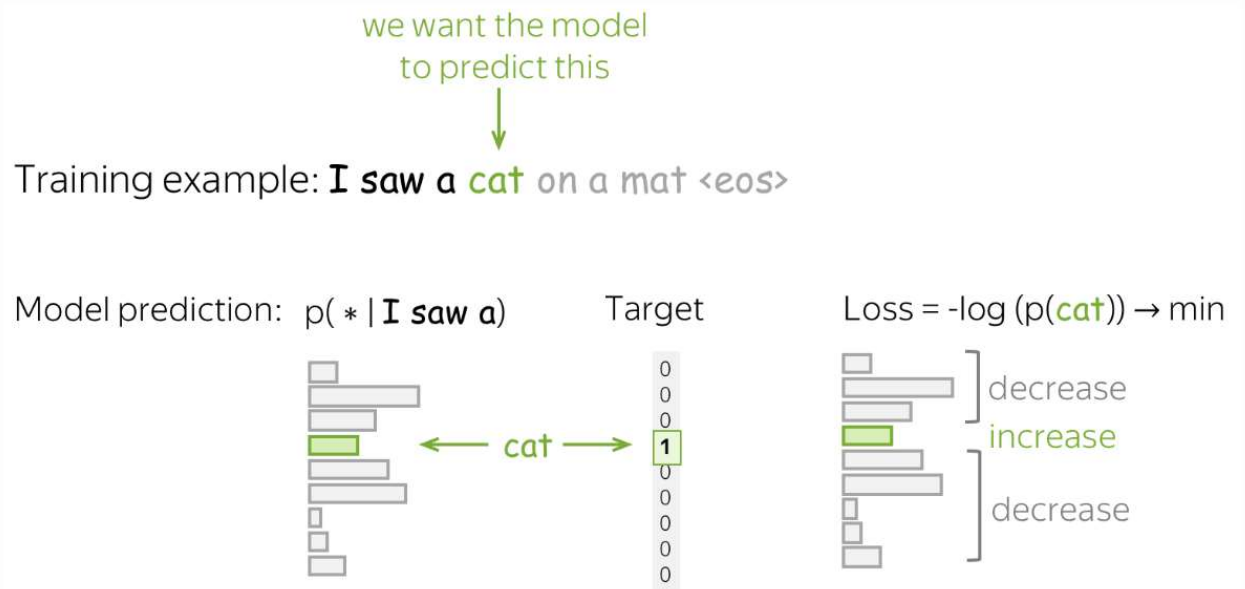


AR Neural Language Models



Loss

- Classify next tokens' index
 - => cross-entropy loss



https://lena-voita.github.io/nlp_course/language_modeling.html#intro

- => maximize text's log-likelihood

$$\max \prod_i p(x_i | x_{1:i-1}) = \min \left(- \sum_i \log p(x_i | x_{1:i-1}) \right) = \min \mathcal{L}(x_{1:L})$$

Tokenizer

- Why?
 - More general than words (eg typos)
 - Shorter sequences than with characters
- Idea: tokens as common subsequences (~3 letters)
- Eg: Byte Pair Encoding (BPE). Train steps:
 1. Take large corpus of text

tokenizer:
text to token
index

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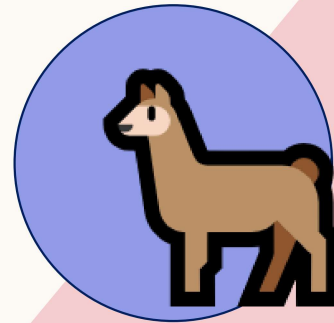
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Overview

Pretraining -> GPT3

- Task & loss
- Evaluation

Post-training -> ChatGPT

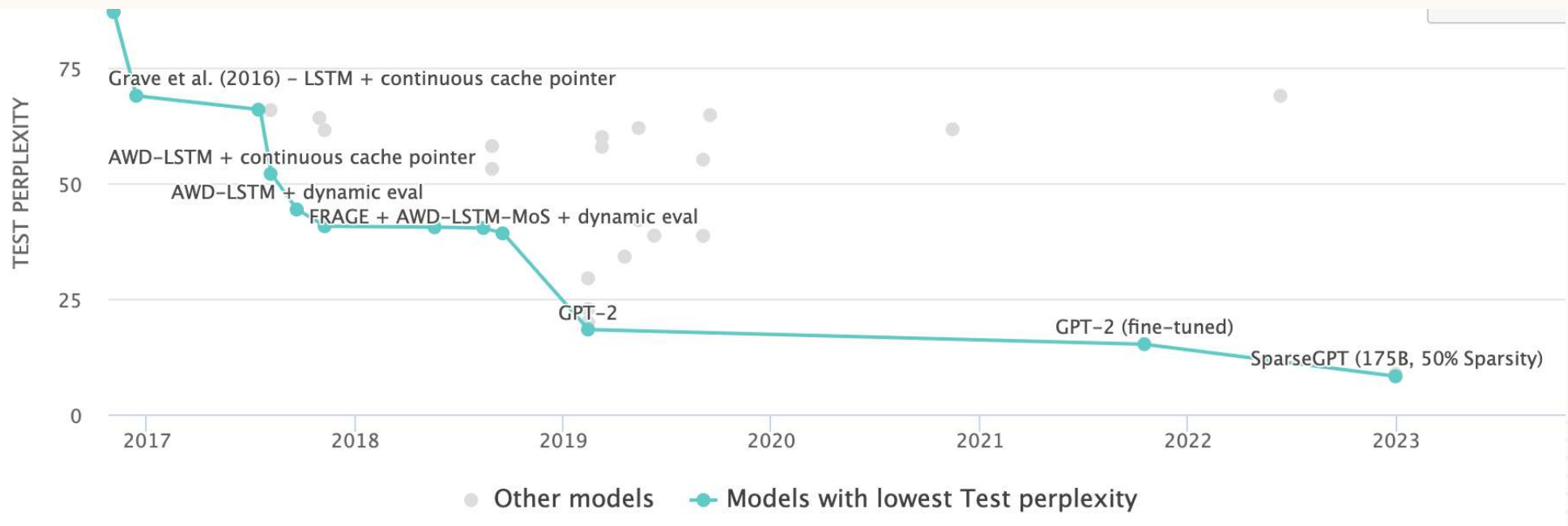


LLM evaluation: Perplexity

- Idea: validation loss
- To be more interpretable: use **perplexity**
 - avg per token (~independent of length)
 - Exponentiate => units independent of log base
- Perplexity: between 1 and |Vocab|
 - Intuition: number of tokens that you are hesitating between

$$PPL(x_{1:L}) = 2^{\frac{1}{L} \mathcal{L}(x_{1:L})} = \prod p(x_i | x_{1:i-1})^{-1/L}$$

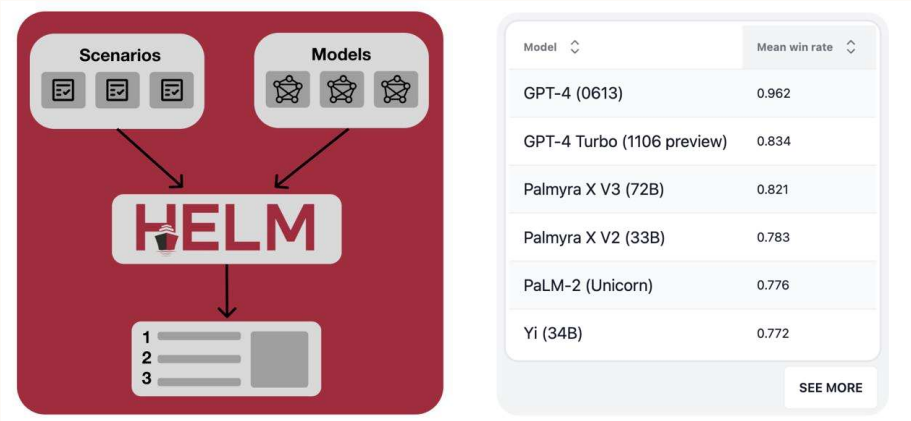
LLM evaluation: Perplexity



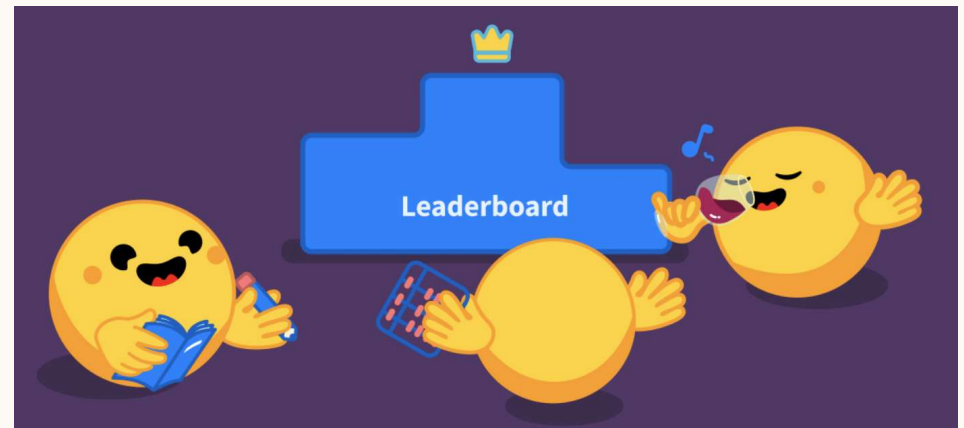
Between 2017-2023, models went from "hesitating" between ~70 tokens to <10 tokens
 Perplexity not used anymore for academic benchmark but still important for development

LLM Evaluation: agg. std NLP benchmarks

Holistic evaluation of language models (HELM)



Huggingface open LLM leaderboard



collect many automatically evaluable
benchmarks, evaluate across them

LLM Evaluation: agg. std NLP benchmarks

- Mix of things that can be “easily” evaluated
- Typically there is “gold” answer
=> you likelihood of LLM to predict that vs other options

HELM-lite
[Liang+ 2022]

Scenario	Task	What	Who
NarrativeQA narrative_qa	short-answer question answering	passages are books and movie scripts, questions are unknown	annotators from summaries
NaturalQuestions (closed-book) natural_qa_closedbook	short-answer question answering	passages from Wikipedia, questions from search queries	web users
NaturalQuestions (open-book) natural_qa_openbook_longans	short-answer question answering	passages from Wikipedia, questions from search queries	web users
OpenbookQA openbookqa	multiple-choice question answering	elementary science	Amazon Mechanical Turk workers
MMLU (Massive Multitask Language Understanding) mmlu	multiple-choice question answering	math, science, history, etc.	various online sources
GSM8K (Grade School Math) gsm	numeric answer question answering	grade school math word problems	contractors on Upwork and Surge AI
MATH math_chain_of_thought	numeric answer question answering	math competitions (AMC, AIME, etc.)	problem setters
LegalBench legalbench	multiple-choice question answering	public legal and administrative documents, manually constructed questions	lawyers
MedQA med_qa	multiple-choice question answering	US medical licensing exams	problem setters
WMT 2014 wmt_14	machine translation	multilingual sentences	Europarl, news, Common Crawl, etc.

LLM Evaluation: eg MMLU

- Example: MMLU
- ~Most trusted pretraining benchmark

Astronomy

What is true for a type-Ia supernova?

- A. This type occurs in binary systems.
- B. This type occurs in young galaxies.
- C. This type produces gamma-ray bursts.
- D. This type produces high amounts of X-rays.

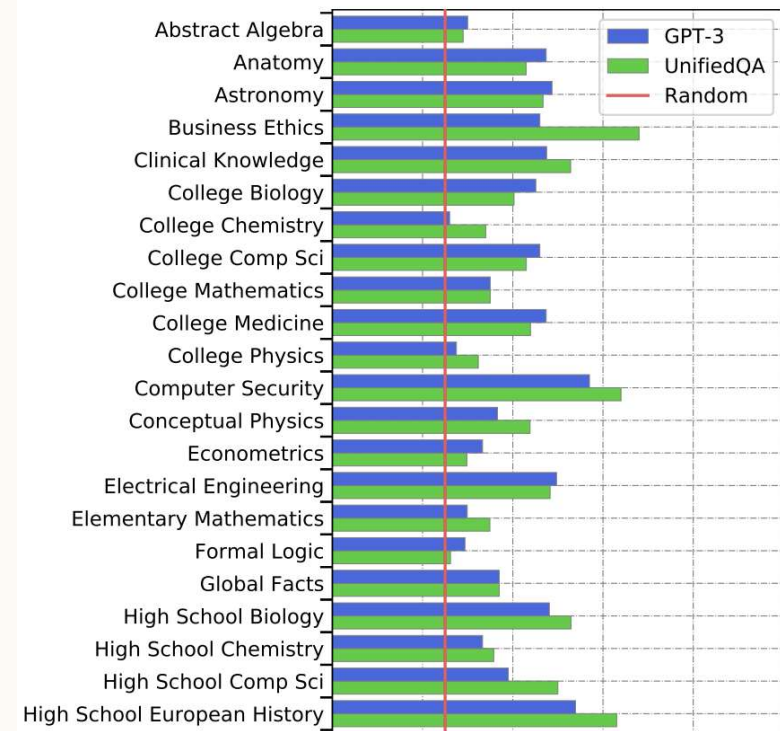
Answer: A

High School Biology

In a population of giraffes, an environmental change occurs that favors individuals that are tallest. As a result, more of the taller individuals are able to obtain nutrients and survive to pass along their genetic information. This is an example of

- A. directional selection.
- B. stabilizing selection.
- C. sexual selection.
- D. disruptive selection

Answer: A



MMLU
[Hendrycks+ 2020]

Evaluation: challenges

- Sensitivity to prompting/inconsistencies

	MMLU (HELM)	MMLU (Harness)	MMLU (Original)
llama-65b	0.637	0.488	0.636
tiiuae/falcon-40b	0.571	0.527	0.558
llama-30b	0.583	0.457	0.584
EleutherAI/gpt-neox-20b	0.256	0.333	0.262
llama-13b	0.471	0.377	0.47
llama-7b	0.339	0.342	0.351
tiiuae/falcon-7b	0.278	0.35	0.254

Evaluation: challenges

- Sensitivity to prompting/inconsistencies
- Train & test contamination (~not important for development)


Horace He
 @cHHillee

I suspect GPT-4's performance is influenced by data contamination, at least on Codeforces.

Of the easiest problems on Codeforces, it solved 10/10 pre-2021 problems and 0/10 recent problems.

This strongly points to contamination.

1/4

g's Race	implementation, math	🚩	🌟	greedy, implementation	🚩	🌟
nd Chocolate	implementation, math	🚩	🌟	cat?	implementation, strings	🚩
triangle!	brute force, geometry, math	🚩	🌟	Actions	data structures, greedy, implementation, math	🚩
	greedy, implementation, math	🚩	🌟	Interview Problem	brute force, implementation, strings	🚩


Susan Zhang ✓
 @suchenzang

I think Phi-1.5 trained on the benchmarks. Particularly, GSM8K.


Susan Zhang ✓ @suchenzang · Sep 12

Let's take [github.com/openai/grade-s...](https://github.com/openai/grade-school-math)

If you truncate and feed this question into Phi-1.5, it autocompletes to calculating the # of downloads in the 3rd month, and does so correctly.

Change the number a bit, and it answers correctly as well.

1/🤖

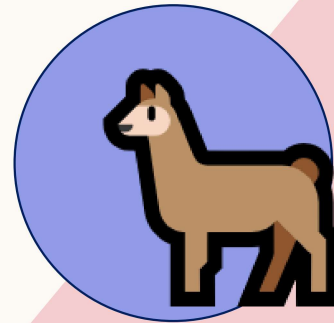



Overview

Pretraining -> GPT3

- Task & loss
- Evaluation
- Data

Post-training -> ChatGPT



Data

- Idea: use all
- Note: inter
 - Down
 - Text e
 - Filter
 - Dedu
 - Heur
 - Mode
 - Data
- Also: lr and

```

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<![endif]-->
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</script>
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</a>
</div>
<div id="menu">
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<input class="inp_text" type="text" name="query" id="query" value="" />
<a href="#" onclick="do_search(false);return false;" class="input_href">

</a>
</form>
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</li>
<li>
<a href="http://www.smartcode.com/db/allrootandsubcats.php">Categories</a>
</li>
<li>
<a href="http://www.smartcode.com/db/new.php">New</a>
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<a href="http://www.smartcode.com/db/top.php">Popular</a>
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h1 strong { float: left; }
div.pager { font-size: 11px; float: right; padding-top: 5px; }
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5<br>
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</td>
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<td>
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</tr>
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1
<a href="/p2.html">2</a>
<a href="/p3.html">3</a>
&nbsp;  
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&nbsp;  
<a href="http://000-084.smartcode.com/macosex.html">Mac</a>
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Download free 000-084 questions and answers. 000-084 exam questions are ultimate...&nbsp;&nbsp;&nbsp;
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Pass-Guaranteed is the leader in IT Certifications that offers a 100% Money Back...&nbsp;&nbsp;&nbsp;
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</p>
</div>
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```

ys same

dirty toks)

g scaling

Data

- Collected from various sources
- Lot of different types of data
 - HTML
 - HTML
- A lot of different types of data
 - ()
- Common
 - ()
 - 5
- Closed

Composition of the Pile by Category

